IT3081 Statistical Modelling

Customer Retention and Future Value

Identifying which customers are likely to return and how much future revenue they may generate.

2026-DS-27 | Customer Lifetime Value Analysis | **Denipitiyage C.N. , Madhusanka K.B.D. , Egalla J.I. , Nanayakkara S.N.N.**

Presentation Roadmap

Four sections connect the online retail problem to evidence-based recommendations.

- Business problem and research context
- Data analysis and predictive modelling
- Evaluation of alternative statistical approaches
- Proposed solution and recommendations

15 min

Customer Value & Retention Challenge

Retail context, business problem and key stakeholders

1. Industry and Dataset Context

UK-based online non-store retailer.

- Transaction-based online sales
- **1.07M+** transaction records.
- **53,628** invoices
- **5,942** identifiable customers
- Customers from **43** countries
- Data period: **Dec 2009 – Dec 2011**

2. Key Stakeholders

- Senior Management
- Sales /Finance
- Marketing / CRM
- Customers

Main Business Problem.
Historical transactions show what customers purchased, but management cannot easily determine who will return or how much future revenue they may generate.

WHY IT MATTERS
Management needs a structured way to identify customers with future value.

CORE BUSINESS QUESTION
Which customers are likely to return, and how much future revenue will they generate?

Consultancy Objectives

Evidence-based customer value framework

1 Predict Customer Return

Logistic Regression
Estimate the probability that a customer will return.

2 Predict Future Revenue

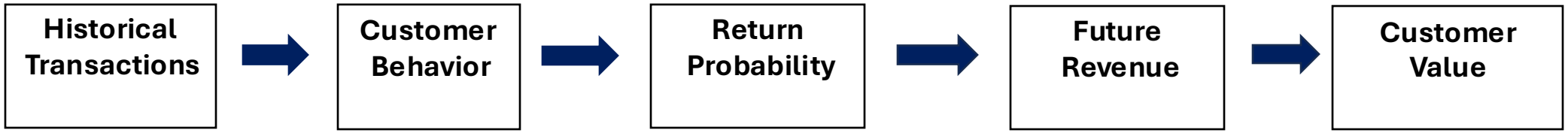
Multiple Linear Regression
Estimate expected future customer revenue.

3 Estimate Customer Value (Proxy)

Combine both predictions:
$\widehat{CLV} = \hat{P}(\text{Return}) \times \hat{R}(\text{Future Revenue})$

4 Support Business Decisions

Identify valuable and at-risk customers and prioritize retention and marketing resources.



Task 2A: Customer Value Modelling Research Landscape

Evidence from 18 peer-reviewed journal studies, including eight published during 2022–2026

Approach	Strength	Limitation
RFM and BG/NBD	Explainable and suitable for non-contractual retail	Relies on purchasing and stationarity assumptions
Logistic and conditional regression	Separates purchase probability from future spending	May miss nonlinear relationships
Boosted and neural models	Capture complex behaviour and temporal patterns	Require more tuning and reduce transparency
Operational CLV systems	Support customer targeting across multiple brands	Require regular updating, calibration and maintenance

No single model performs best in every retail setting. The appropriate method depends on data availability, customer behaviour, forecast horizon and operational requirements.

Task 2B: Research Gap and Analytical Response

Addressing limitations in customer-value definition, data handling and model evaluation

Research gap	Why it matters
Value definition	Short-term revenue is sometimes incorrectly described as lifetime profit
Data treatment	Returns, cancellations and wholesale orders can distort customer value
Model evaluation	Random data splits can leak future information into model training
Generalisability	Results from other retailers may not transfer directly to this dataset

Logistic regression predicts repeat-purchase probability, while linear regression estimates revenue if the customer returns. Their combination provides an interpretable estimate of future customer value.

Task 3A: Dataset Selection & Understanding

Why the selected dataset is suitable for the consultancy problem

Dataset Source & Accessibility

- UCI Machine Learning Repository
- Publicly accessible dataset
- Online Retail II transaction data
- Suitable for reproducible statistical analysis

Key Variables

- CustomerID – identifies customers
- InvoiceDate – transaction timing
- Quantity – units purchased
- UnitPrice – price per unit
- InvoiceNo / Country – transaction and customer context

Target Outcomes

- Customer Return – whether a customer purchases again
- Future Revenue – revenue generated in the future period
- These outcomes support the customer value / retention problem

Why Suitable for the Consultancy Problem

- Provides historical customer purchasing behavior
- Supports repeat-purchase / retention analysis
- Contains variables for future revenue prediction
- Large number of observations supports statistical modelling
- Results can support retention and marketing decisions

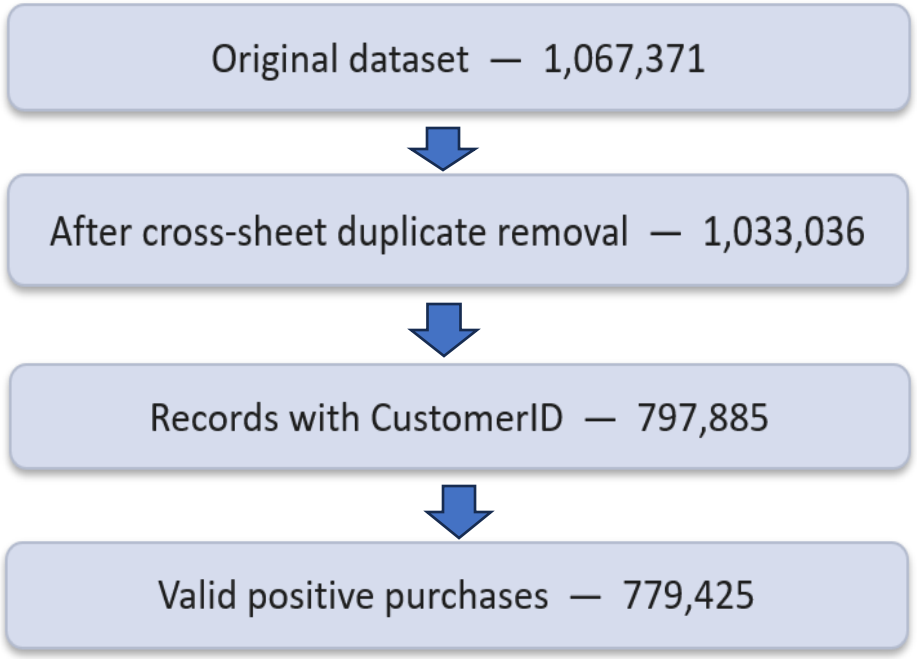
Task 3B: Data Quality Assessment & Cleaning

Identifying and handling records that could distort customer-level analysis.

Issue	Result
Missing CustomerID	243,007 (22.77%)
Exact duplicate rows	12,133
Cancellation / return rows	19,494
Negative quantity rows	22,950
Zero-price rows	6,202
Negative-price rows	5

Outliers: Large values were retained when they could represent legitimate wholesale activity; log transformations and robust statistical methods were used.

Cleaning & Customer Identification



Final customer-level analytical dataset
4,266 customers | 32 variables

Task 3C: Descriptive Statistics

Customer-level analysis using the feature-engineered dataset

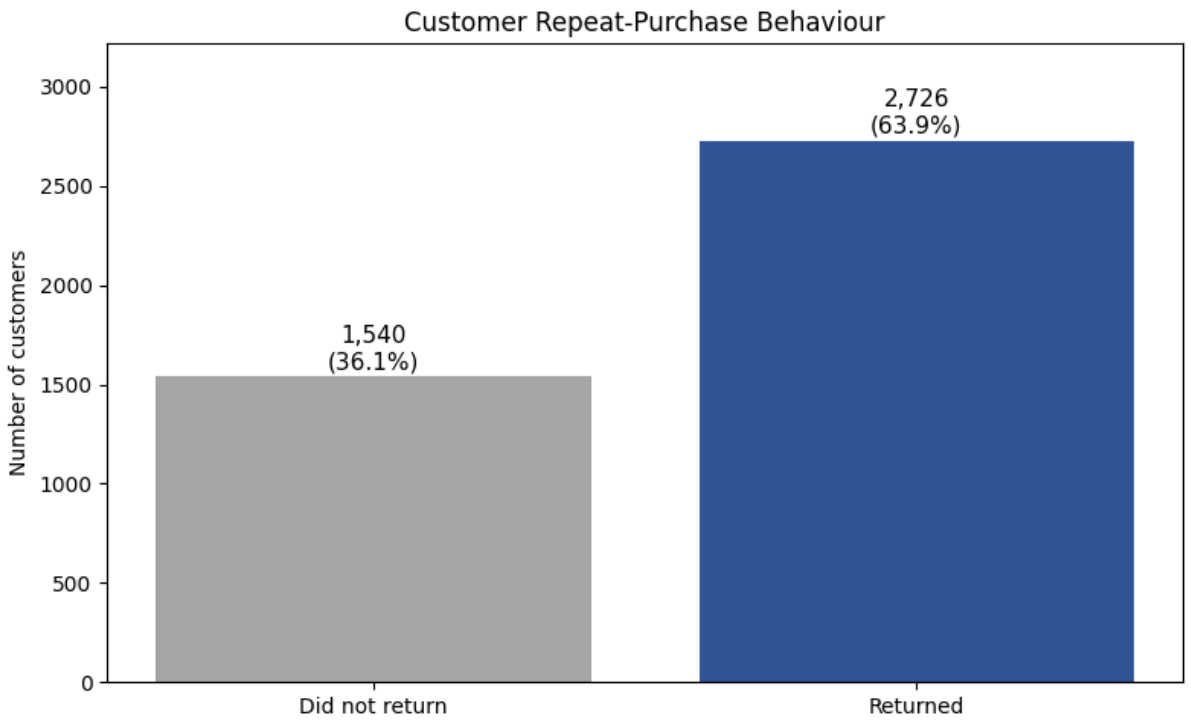
Customer measure	Result
Customer-level observations	4,266 customers
Purchase frequency	2 orders (median)
Monetary value	681.81 (median)
Recency	47 days (median)
Repeat-purchase rate	63.9%
Future revenue	314.88 (median)

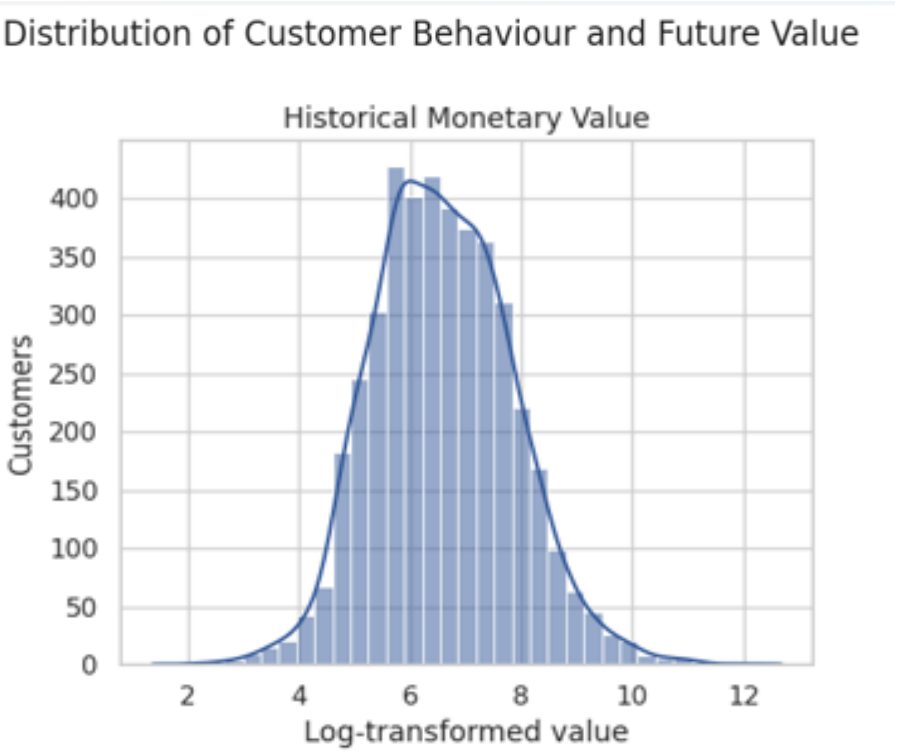
Key observations

- The statistical unit is the customer, not the individual transaction.
- The typical customer placed 2 orders and spent 681.81.
- Median recency was 47 days.
- 63.9% of customers made a repeat purchase.
- Median future revenue was 314.88

Task 3D: Key Graphs & Customer Behavior Patterns

Simple visuals showing the main patterns in the customer-level data.





Repeat-purchase behavior

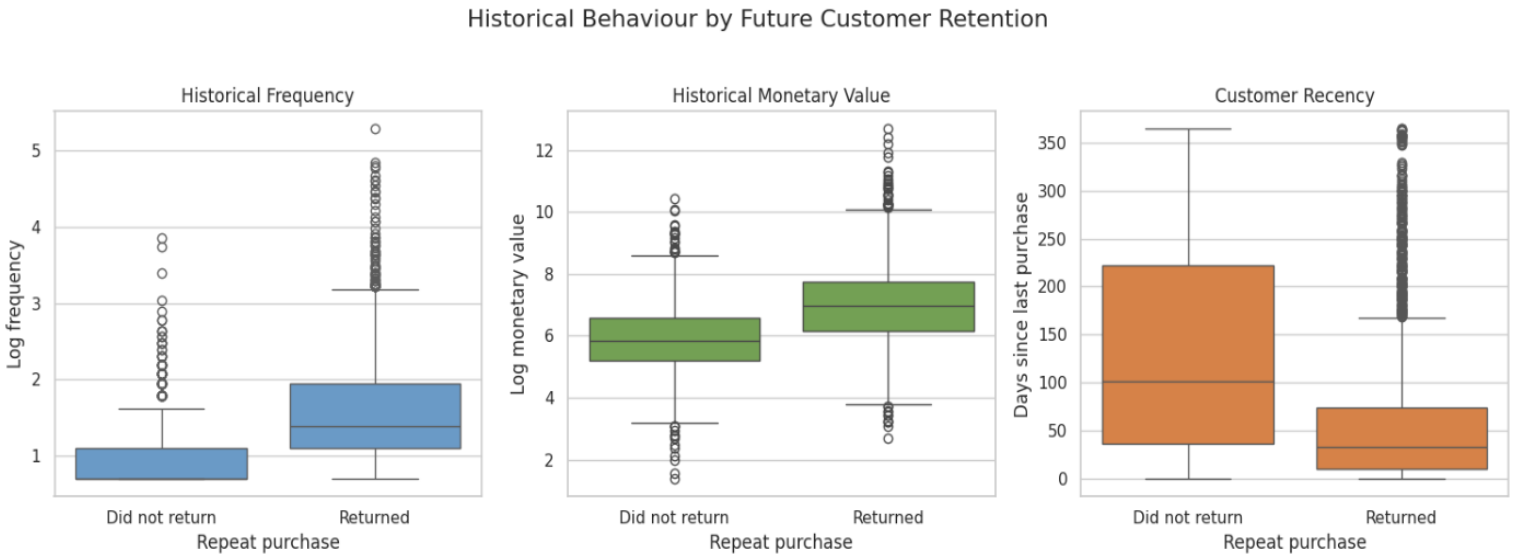
2,726 customers (63.9%) returned; 1,540 (36.1%) did not return.

Historical monetary value

Log transformation reduces right-skewness and makes the spending distribution easier to analyze.

Task 3E: Initial Business Insights

What management can learn from customer- level behavior.



Initial Insights

- Customer behavior is highly uneven.
- Returning customers had higher historical frequency: median 3 vs 1 order.
- Returning customers had higher monetary value: median 1,061.76 vs 346.53.
- Returning customers had lower recency: median 33 vs 102 days.

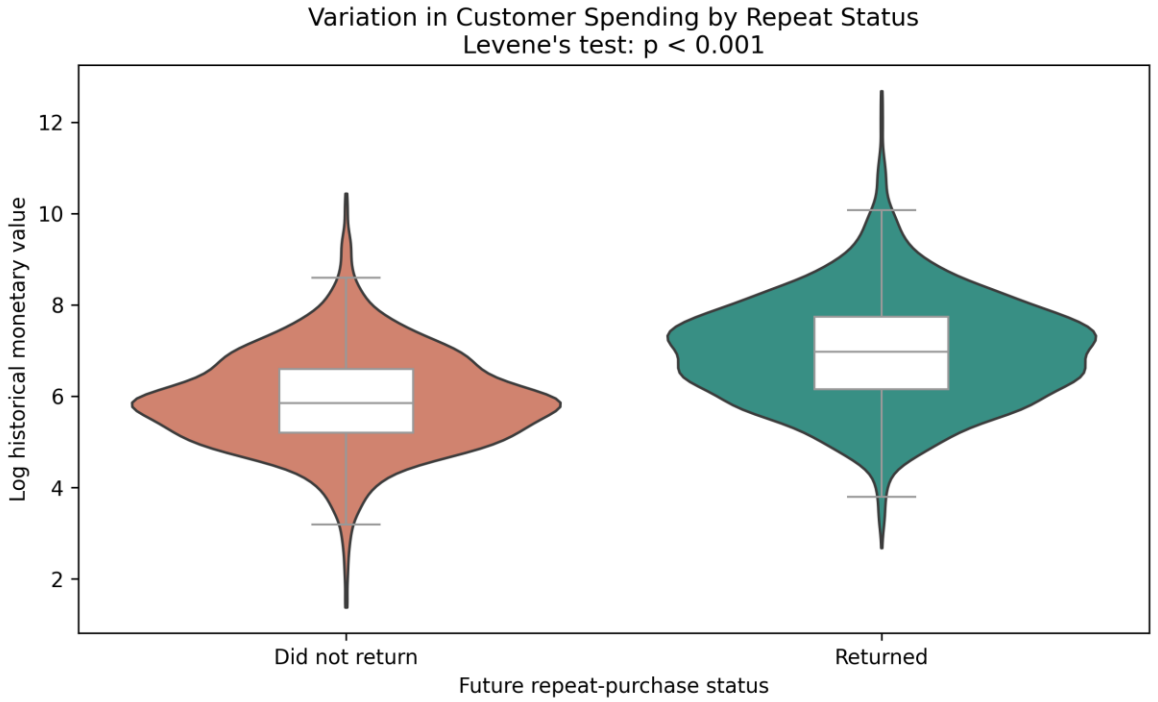
What Management Can Learn

- Frequency, monetary value and recency can support customer segmentation.
- Customer behavior can help identify different retention groups.
- Repeat purchase and future revenue should be analyzed as separate outcomes.

These patterns support customer segmentation and the separate prediction of repeat purchase and future revenue.

Task 4A: Variability in Historical Customer Value

Testing whether spending variability differs between customers who returned and those who did not.



1. **Variance comparison - Levene's test**

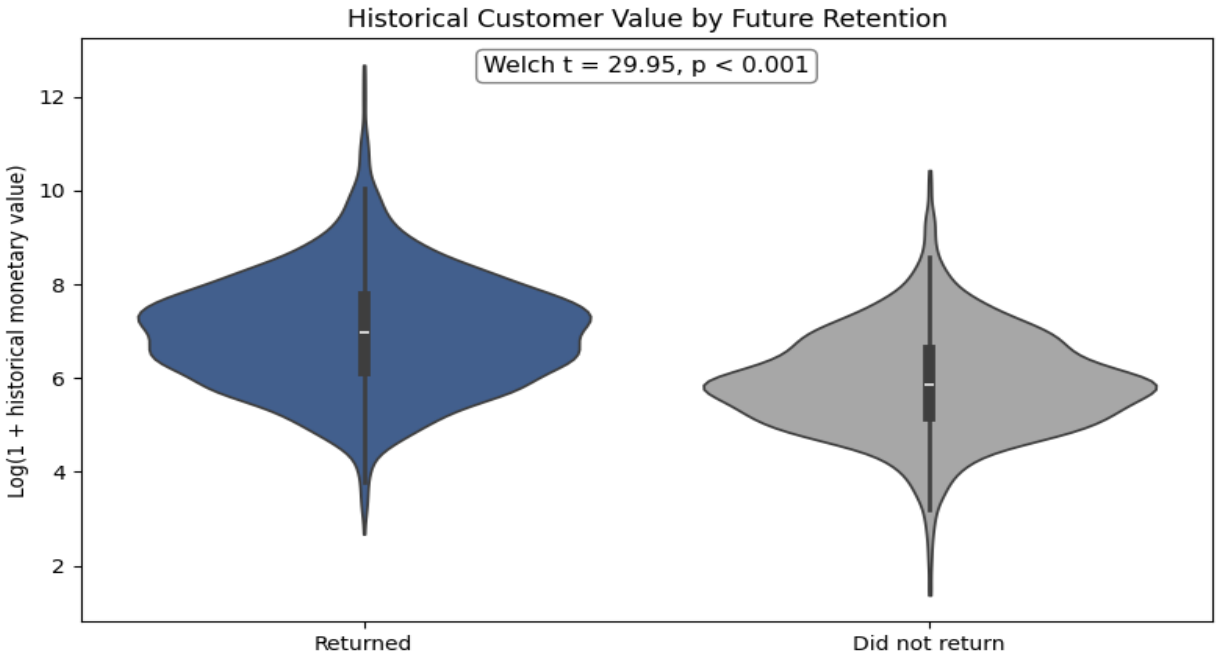
$$H_0: \sigma^2_{\text{returned}} = \sigma^2_{\text{not returned}}$$
$$W = 32.27, p < 0.001$$

Decision: Variances differ significantly.

Historical spending varies more among returning customers. Because the equal-variance assumption is violated, Welch's t-test is used for the mean comparison.

Task 4B: Mean Historical Value by Future Retention

Testing whether average historical customer value differs between the two retention groups.



Welch t-test

$$H_0: \mu_{\text{returned}} = \mu_{\text{not returned}}$$
$$H_1: \mu_{\text{returned}} \neq \mu_{\text{not returned}}$$

Result

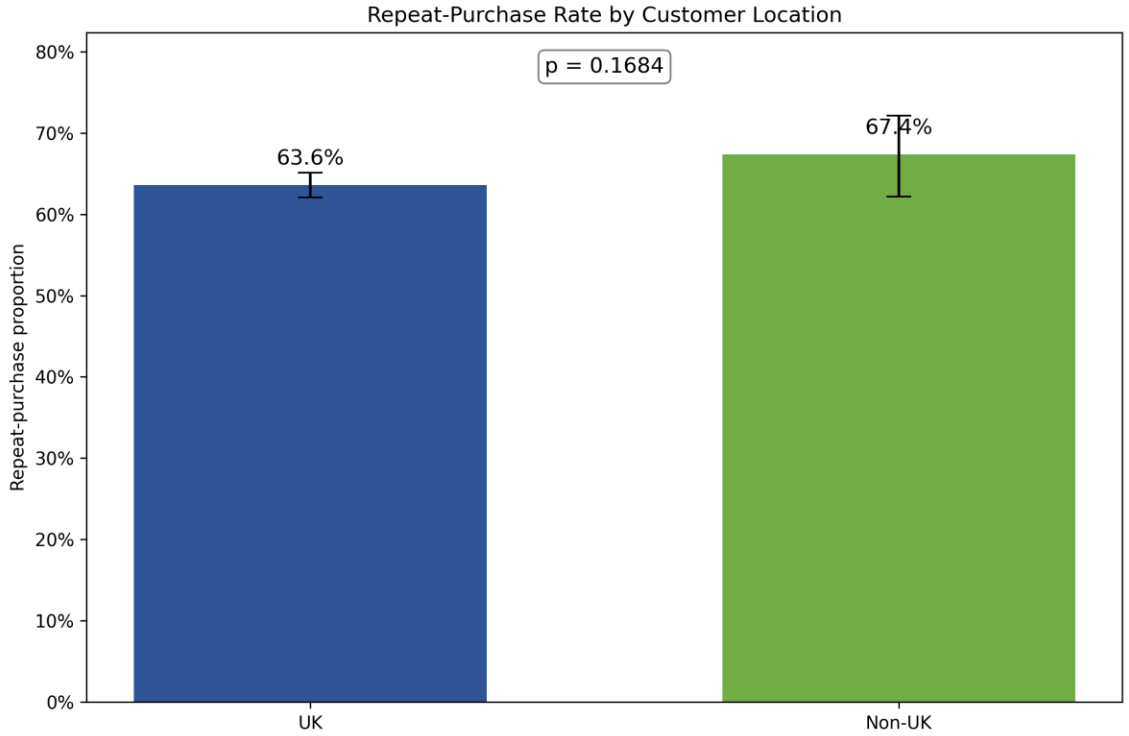
$$t = 29.95, p < 0.001$$
$$\alpha = 0.05$$

Decision: Reject H0

Returning customers had significantly higher historical spending, showing that past customer value is useful for retention targeting and future-value prediction.

Task 4C: Customer Location and Repeat-Purchase Rate

Testing whether repeat-purchase proportions differ between UK and non-UK customers.



Two-proportion z-test

$$H_0: p_{\text{UK}} = p_{\text{NonUK}}$$
$$H_1: p_{\text{UK}} \neq p_{\text{NonUK}}$$

Results

$$z = -1.38, p = 0.1684$$
$$\alpha = 0.05$$

Decision: Fail to reject H0

Repeat-purchase rates did not differ significantly between UK and non-UK customers ($p = 0.1684$); therefore, retention targeting should rely on customer behaviour rather than location alone.

Predictive Statistical Modelling

Building a two-stage statistical framework to estimate customer retention probability and future revenue value.

1 Model Selection & Justification

Logistic Regression selected for repeat-purchase prediction; Linear Regression selected for future revenue estimation.

2 Alternative Models Compared

Logistic Regression and MLR gave the expected results. Alternatives would add complexity without improving the CLV, so we kept the simplest pipeline that works.

3 Model Performance & Assumptions

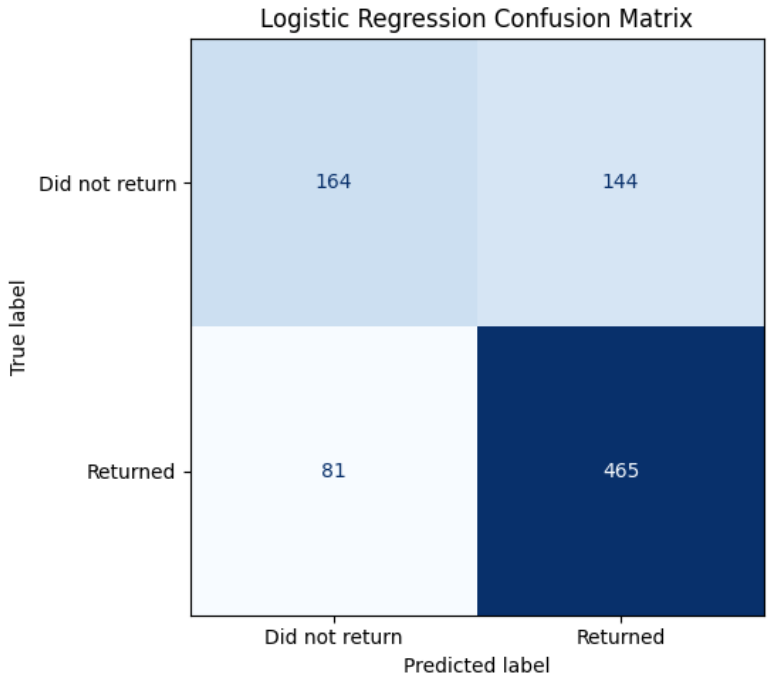
Evaluated on a held-out test set using Accuracy, Precision, Recall, F1-score, ROC-AUC, R^2 and RMSE.

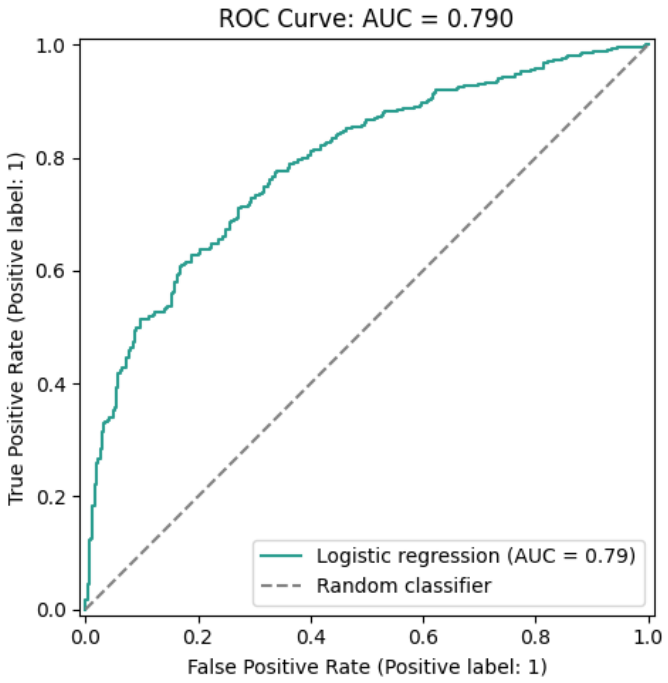
4 Recommended Model for Organizational Use

Integrated CLV framework:
CLV = P(Return) × E(Future Revenue | Return)

Task 5A: Logistic Regression for Repeat-Purchase Prediction

Estimating the probability that a customer returns during the future observation period.





Test performance

- Accuracy: **73.65%**
- Precision: **76.35%**
- Recall: **85.16%**
- F1-score: **80.52%**
- ROC-AUC: **0.790**

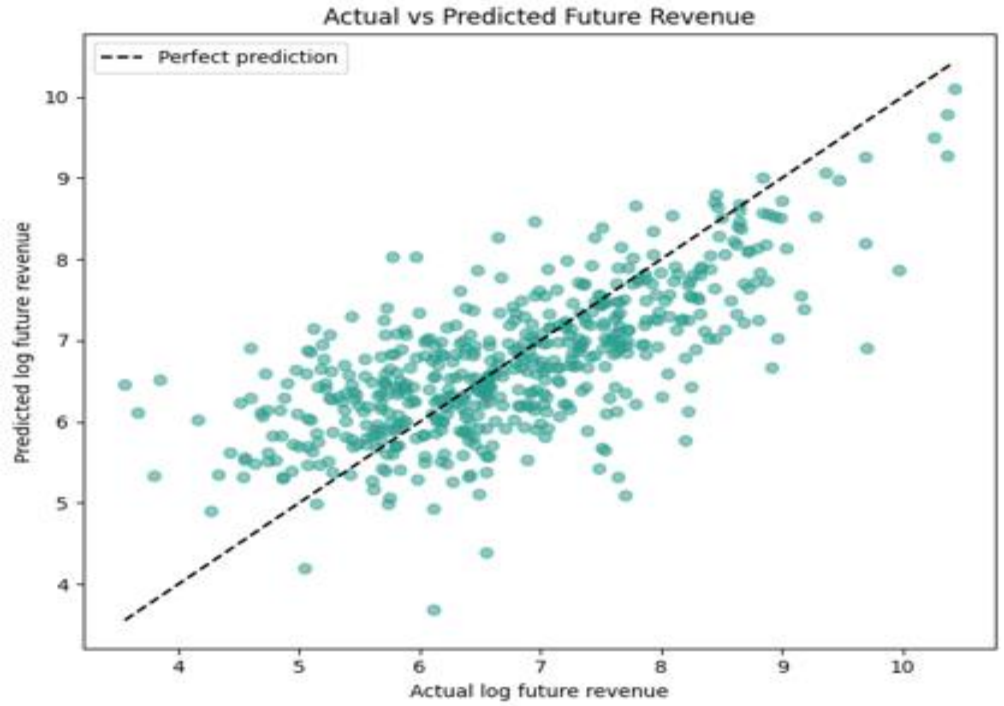
Model stability

Five-fold CV AUC = **0.786 ± 0.008**

Logistic regression beat the 63.9% majority-class baseline and identified 85% of returning customers, but found only 53% of non-returners.

Task 5B: Multiple Linear Regression for Future Revenue

Predicting log future revenue for returning customers, then converting back to revenue with a smearing correction.



Predictive performance

- Revenue-scale R^2: **0.681**
- Log-scale R^2: **0.472**
- MAE: **1,048.07**
- RMSE: **1,943.17**

Coefficient evidence

- Positive: **frequency, average order value**
- Negative: **recency, product diversity**
- Removed: **monetary value (redundant)**

The corrected model explains 68.1% of revenue variation; dropping the redundant predictor gave clearer coefficients at little cost in accuracy.

Task 5C: Integrated CLV Model & Customer Segments

Combining both models into a single customer value indicator.

THE TWO-STAGE CLV FRAMEWORK

$$CLV_i = \hat{P}(Return_i) \times \hat{E}(Future\ Revenue_i \mid Return_i = 1)$$

$\hat{P}(Return_i)$ — LOGISTIC REGRESSION

$\hat{E}(Future\ Revenue_i \mid Return_i = 1)$ — MULTIPLE LINEAR REGRESSION

CLV SEGMENT RESULTS (QUARTILE-BASED)

Segment	Customers	Mean CLV (£)	Median CLV (£)	Mean Return Probability
Low	1,067	127.42	124.61	0.33
Medium	1,066	422.84	413.00	0.58
High	1,066	1,026.00	1,000.18	0.75
Very High	1,067	5,062.58	2,727.71	0.89

The top half of customers hold 92% of the value - retention spend should focus there

Task 6A: Proposed Retention Experiment

Blocking customers by predicted CLV segment before randomising retention offers.

Random Allocation within CLV Blocks

CLV block	Control	10% Discount	Free Delivery
Low	**356** customers	**356** customers	**355** customers
Medium	**356** customers	**355** customers	**355** customers
High	**356** customers	**355** customers	**355** customers
Very High	**356** customers	**356** customers	**355** customers

Experimental design

- Design : RCBD
- Blocks : 4 CLV segment
- Treatments : control, 10% Discount, Free delivery
- Response :Revenue within 60 days

Planned analysis

Two-way ANOVA (treatment + CLV block)

Randomising within CLV blocks gives balanced groups (355–356 customers per cell), so revenue differences reflect the offer rather than customer value.

Task 6B: CRD vs RCBD Evaluation

Comparing a completely randomised design with a design that blocks on customer value.

Completely Randomised Design (CRD)

Criterion	Rating
Customer variation control	Low
Complexity	Low
CLV suitability	Moderate

Simple to run, but ignores differences between high- and low-value customers.

Randomised Complete Block Design (RCBD)

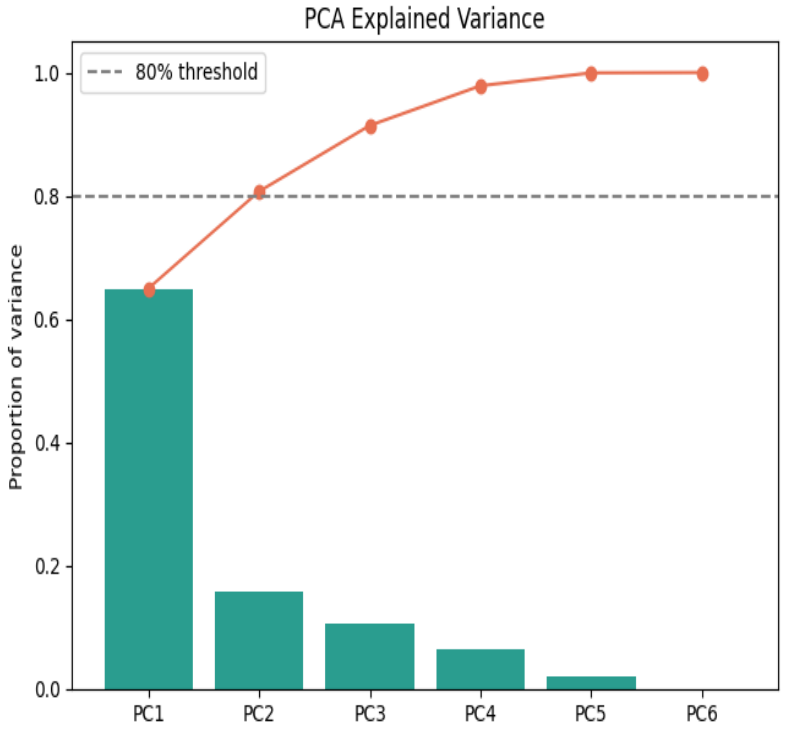
Criterion	Rating
Customer variation control	High
Complexity	Medium
CLV suitability	Strong

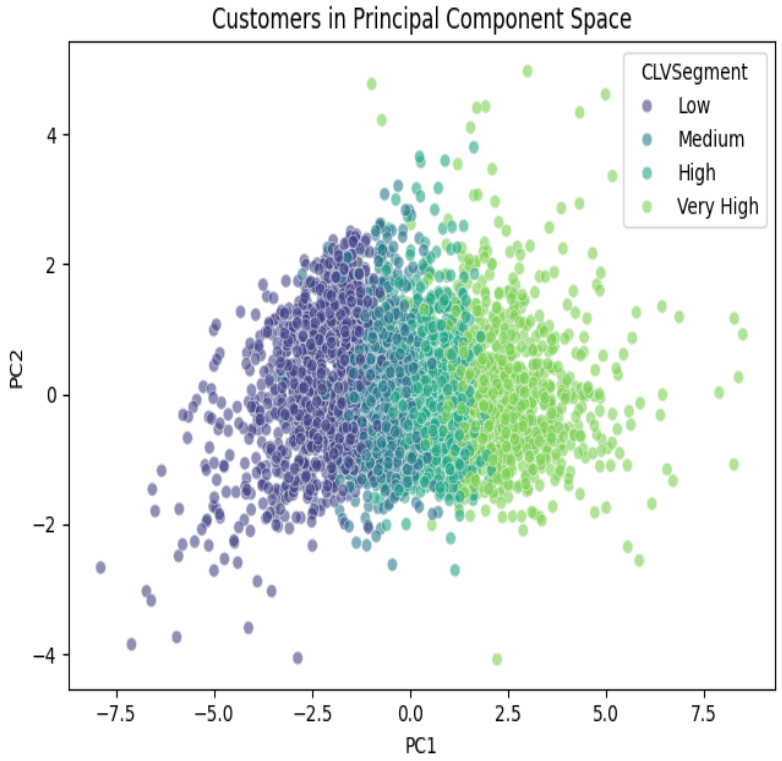
Compares offers within similar-value customers, giving a fairer test of each treatment.

RCBD is preferred because customer value differences can influence retention outcomes.

Task 7A: PCA Results and Interpretation

Evaluating whether dimensionality reduction improves customer behaviour analysis.





VARIANCE BY COMPONENT

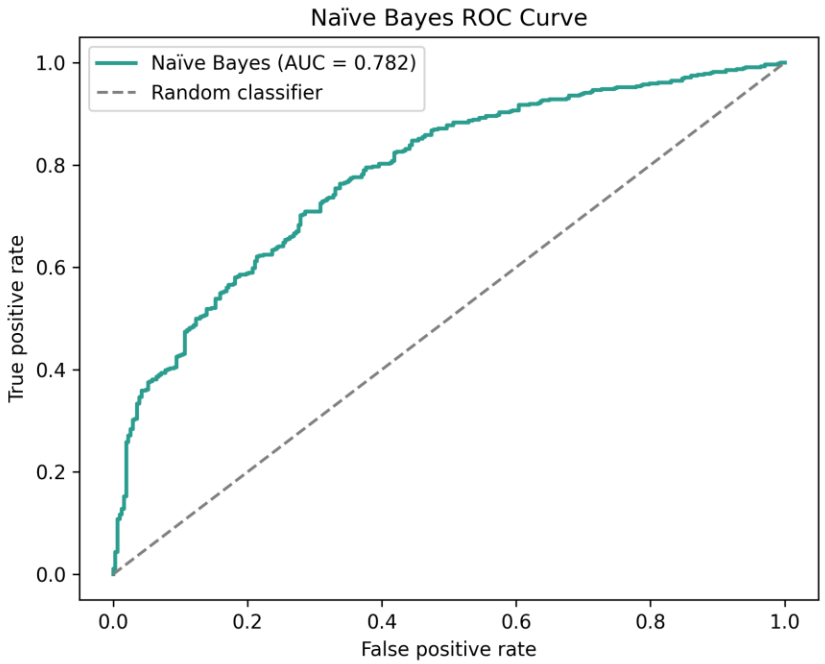
Component	Individual	Cumulative
PC1	64.97%	64.97%
PC2	15.75%	80.72%
PC3	~10.5%*	~91.3%*
PC4	~6.5%*	~97.8%*
PC5	~2.0%*	~99.8%*
PC6	~0.2%*	100.0%*

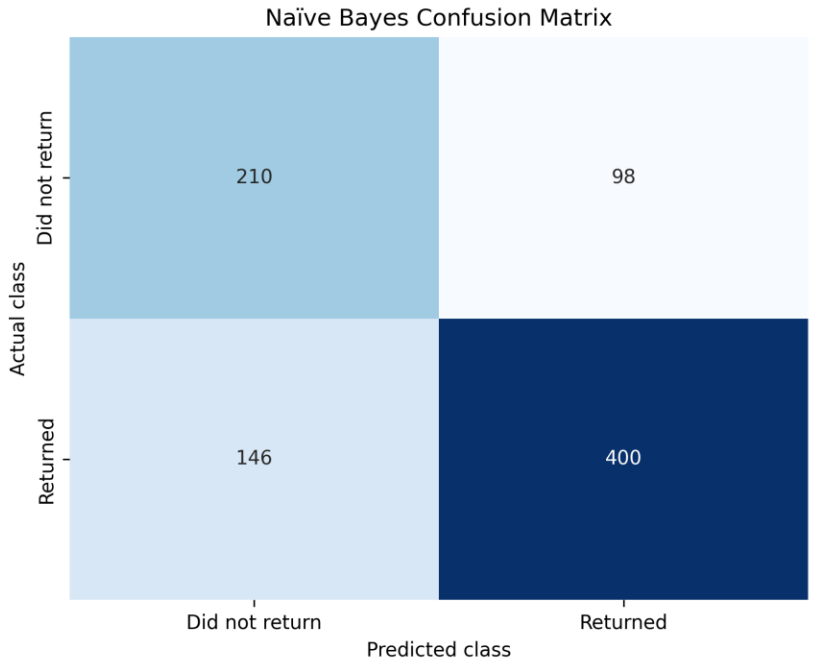
6 features → 2 components = 66.7% fewer dimensions. Two components clear the 80% mark, but reaching 95% needs four — so a 2D summary trades real accuracy for simplicity.

Two components keep 80.72% of the variance, but they lose business meaning, so original variables are preferred for CLV prediction.

Task 8A: Naïve Bayes for Repeat-Purchase Prediction

Evaluating a probabilistic alternative to logistic regression for identifying returning customers.





Test performance

- Accuracy: **71.4%**
- Precision: **80.3%**
- Recall: **73.3%**
- F1-score: **76.6%**
- ROC-AUC: **0.782**

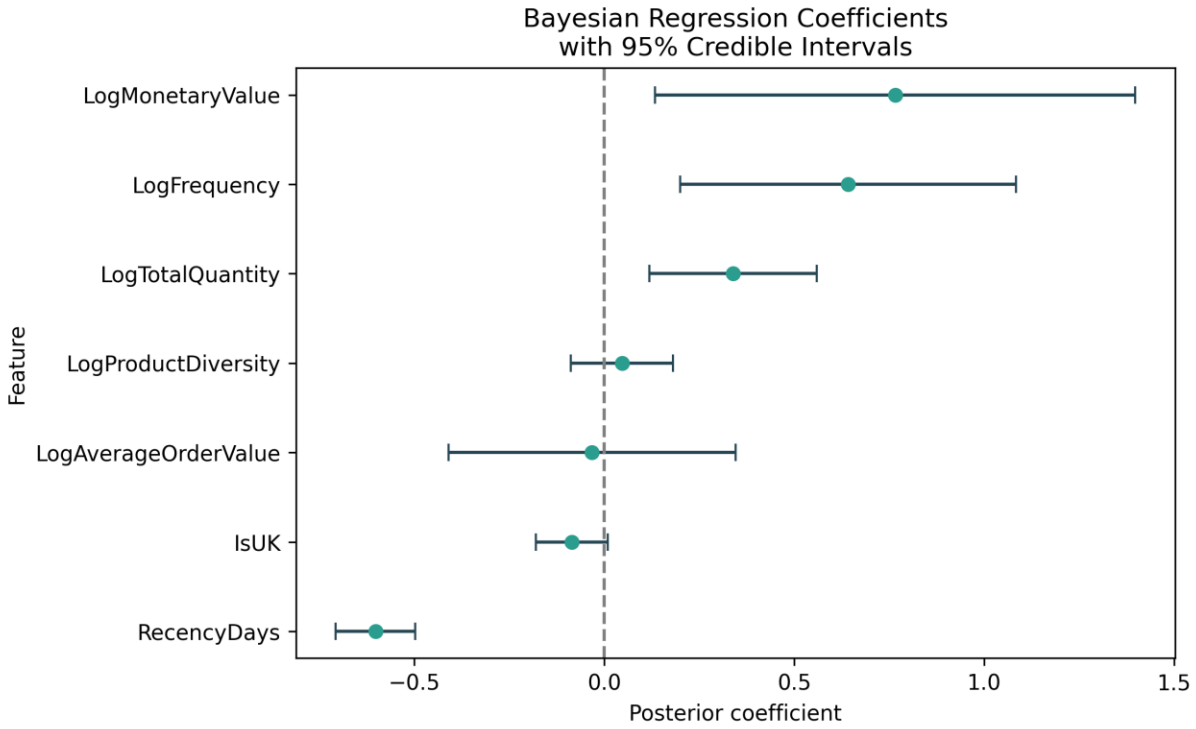
Model stability

Five-fold CV AUC=0.779±0.014

Naïve Bayes provided a stable retention benchmark, but its independence assumption limits its advantage over logistic regression.

Task 8B: Bayesian Regression for Future Revenue

Predicting future customer revenue while quantifying uncertainty in the model coefficients and predictions.



Predictive performance

- Log-scale R^2: **0.387**
- Revenue-scale R^2: **0.591**
- MAE: **1,173**
- Five-fold log R^2: **0.351 ± 0.027**
- 95% interval coverage: **96.4%**

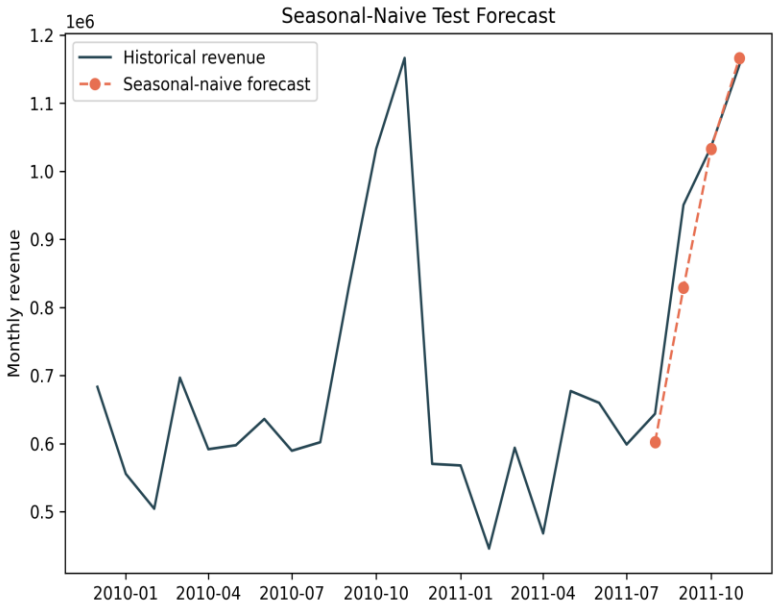
Posterior evidence

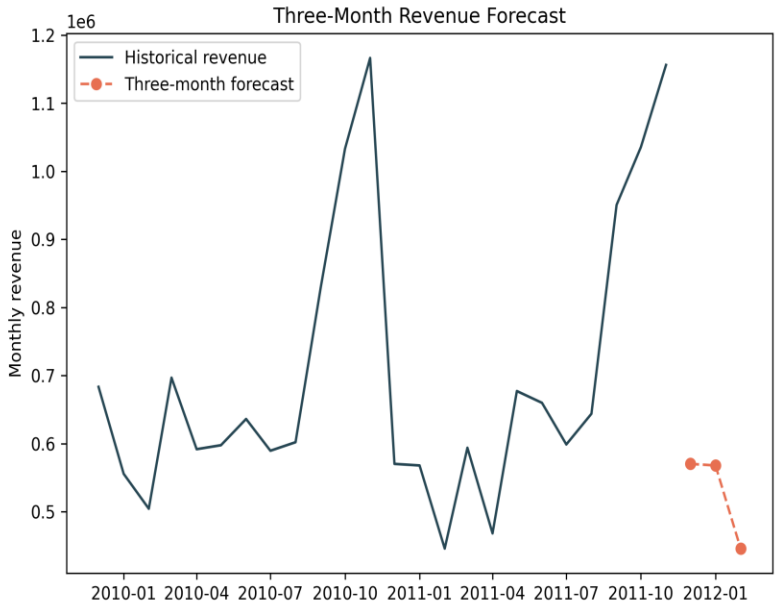
- Positive: frequency, monetary value and quantity
- Negative: recency
- Uncertain: location, average order value and product diversity

Bayesian Ridge explained 59.1% of revenue variation and produced well-calibrated uncertainty intervals, supporting risk-aware customer targeting.

Task 9A: Seasonal Monthly Revenue Forecasting

A seasonal-naïve model captured the annual revenue pattern more accurately than a non-seasonal trend model.





Data and method

- **24 complete months**
- Partial December 2011 data excluded
- Final four months used for testing
- Seasonal-naïve forecast: same month from the previous year

Revenue peaks from September to November and declines after the seasonal peak, supporting short-term inventory, staffing and cash-flow planning.

Task 9B: Forecast Accuracy and Model Selection

Comparing a non-seasonal trend model with a model that preserves the annual revenue pattern.

Non-seasonal Holt model

Metric	Test result
MAE	335,153.80
RMSE	384,962.07
MAPE	32.20%

The model missed the strong annual sales pattern and substantially underestimated peak-period revenue.

Seasonal-naïve model

Metric	Test result
MAE	44,071.89
RMSE	64,548.66
MAPE	5.11%

Using revenue from the corresponding month of the previous year captured the observed seasonal pattern more accurately.

The seasonal-naïve model reduced MAPE from 32.20% to 5.11%, an improvement of approximately 84%, and was selected for the final three-month forecast.

Task 10A: Industry Innovation Proposal – Business Need

Predictive Blindspot

Management cannot easily determine which customers will return or estimate the future revenue they will generate.

Value & Seasonality

There is a critical need to identify high-value customers and anticipate the strong September–November seasonal revenue peaks.

Behavior Over Location

Retention strategies must be optimized away from location-based targeting (which proved statistically insignificant) and focus on actual customer behavior.

Task 10A: Industry Innovation Proposal – Proposed Solution

A two-stage predictive dashboard combining our most effective statistical models

Stage 1: Retention Prediction

Logistic Regression

(Probability of customer return)

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Stage 2: Future Revenue Prediction

Multiple Linear Regression

(**Expected future revenue**)

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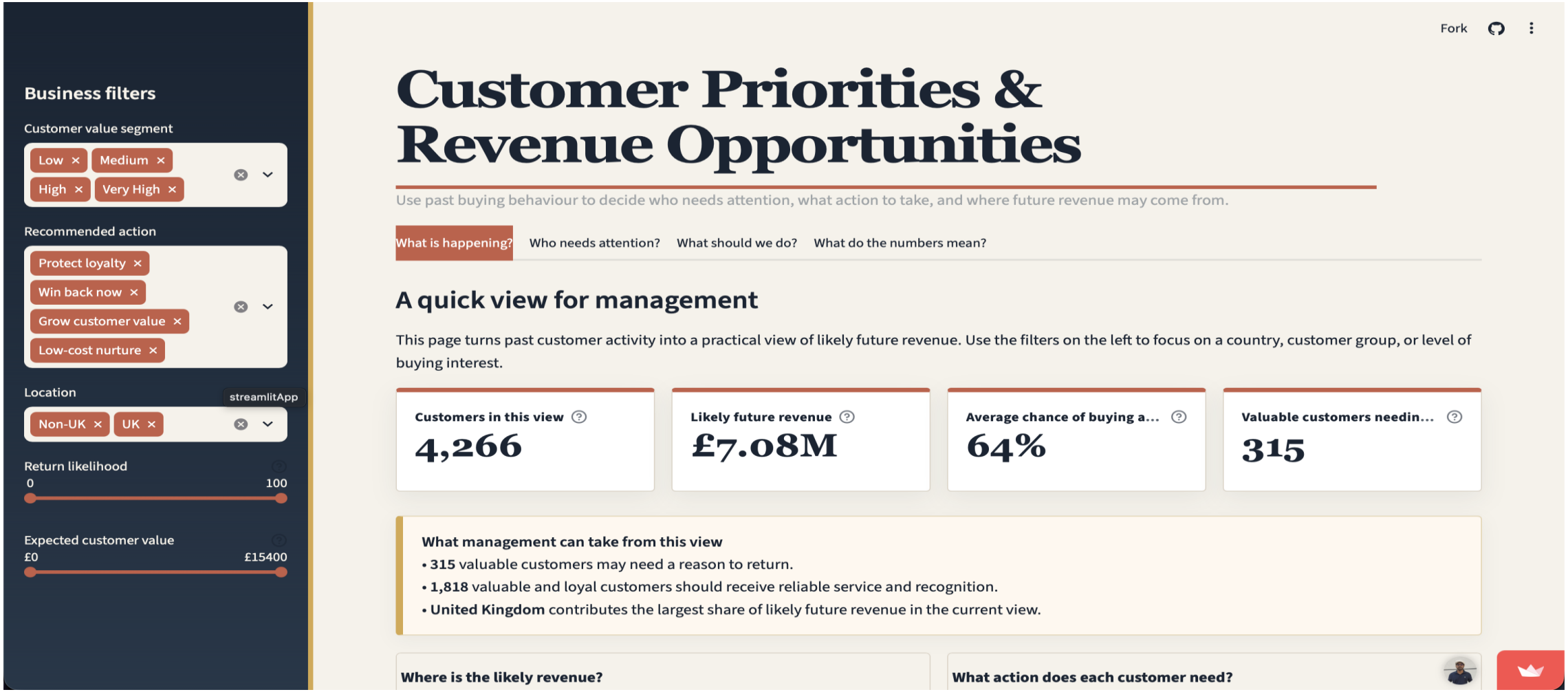
Customer Lifetime Value (CLV) Indicator

The ultimate metric for targeted retention.

CLV = P(Return) × E(Future Revenue | Return)

Task 10B: Proposed Dashboard Display

The dashboard will combine key customer metrics with predictive insights:



Task 10C: Expected Benefits, Challenges & Implementation

THE WHY • POSITIVE OUTCOMES

Expected Benefits

- Proactive churn mitigation and optimized marketing ROI.
- Improved cash-flow planning via Seasonal-Naïve forecasts.

Required Resources

- Cloud database integration.
- CRM marketing software.
- Dedicated data analytics personnel.

THE HOW • REQUIREMENTS & HURDLES

Organizational Impact

- Transforms the company from reactive historical reporting to proactive, predictive retention.

Implementation Challenges

- Maintaining data quality across ongoing transactions.
- Training marketing teams to interpret model predictions and prediction uncertainty.

Task 12A: Strategic & Operational Recommendations

Strategic Recommendations

- **Behavior-Based Targeting**
Cease location-based (UK vs. Non-UK) retention campaigns, as statistical evidence (p = 0.1684) proves location does not drive repeat purchases.

- **Transparent Models**
Avoid black-box dimensionality reduction (PCA). Maintain original variables (Recency, Frequency, Monetary) to ensure business interpretation remains clear.

Operational Recommendations

- **Automate Retention Actions**
Deploy the Logistic Regression model (AUC = 0.790) to automatically flag high-value customers showing declining recency.

- **Seasonal Resource Allocation**
Utilize the Seasonal-Naïve forecast model (which reduced MAPE by ~84%) to scale inventory, staffing, and marketing budgets for the September–November revenue peaks.

Evidence-led recommendations • p = 0.1684 • 84% MAPE reduction

Task 12B: Risk, Ethics & Future Steps

Risk Management

- **Model Uncertainty**
  Utilize the 95% credible intervals from the Bayesian Ridge model to avoid overcommitting marketing spend.
- **Model Decay**
  Continuously monitor the logistic regression baseline (identifying 85% of returners) and retrain annually.

Ethical Considerations

- **Data Privacy**
  Ensure rigorous compliance with GDPR/data protection regulations regarding the storage and processing of historical transaction data.
- **Transparent Profiling**
  Avoid algorithmic bias by ensuring retention models rely strictly on purchasing behavior, not demographic profiling.

Future Research Opportunities

- **Product-Level Analysis**
  Expand the dataset to include product categories to understand if specific items drive higher retention.
- **Experimental Design**
  Implement a CRD or RCBD A/B testing framework to measure the true causal lift of specific marketing emails.

Thank You

2026-DS-27 | Customer Lifetime Value Analysis

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